Investigating AI Adoption in Construction Companies

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Abstract

This report examines how artificial intelligence (AI) is being used or considered in two companies connected to Architecture, Engineering, and Construction (AEC)-related. The study focuses on Gosalia Concrete Constructors, Inc., a concrete construction company, and Estudios y Consultorias, S.A.S., a financial consulting company. A qualitative approach was used through phone interviews with two professionals: Juan Romero, Lead Estimator and Diego Rapalino, Financial Analyst. The interviews focused on AI tools, possible uses, benefits, risks, barriers, and future expectations. The findings showed that both companies see AI as a useful and powerful tool, especially for organizing information, summarizing reports, preparing drafts, and supporting decision-making. However, both interviews also showed concerns related to accuracy, confidentiality, trust, training, and the need for human review. The report concludes that AI has strong potential to improve efficiency in AEC-related work, but it should not replace professional judgement. Instead, AI should be used carefully in tasks where it adds value while keeping final decisions in the hands of experienced professionals.

Keywords: artificial intelligence, construction, AEC industry, technology adoption, innovation

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Introduction

Artificial intelligence has quickly become one of the most discussed technologies across professional industries. Companies are looking for ways to use AI to improve productivity, organize large amounts of information, and support better decision-making. Since generative AI tools have become easier to access, more companies are starting to look at how these tools can fit into their normal operations, not only for technical work, but also for management, reporting, estimating, communication, and daily business tasks (McKinsey & Company, 2023).

In the Architecture, Engineering, and Construction (AEC) industry, AI can be useful in many different areas. It can support estimating, scheduling, project management, safety, risk analysis, document review, design review, and financial planning. However, using AI in AEC is not always straightforward because the industry depends heavily on accuracy, experience, and specific conditions for each project. A tool may be able to summarize a contract or review a specification, but someone still needs to check the information and make sure it applies to the actual project. Several studies have found that AI can help construction companies improve efficiency and decision-making, but adoption still has many challenges such as training, cost, trust, data quality, and organizational readiness (Abioye et al., 2021; Regona et al., 2022).

This report focuses on two companies connected to AEC-related work. The first one is Gosalia Concrete Constructors, Inc., a concrete construction company that specializes in roadway barrier walls, concrete pavement, concrete slab replacement, and other concrete-related work. For this company, AI could be helpful for estimating support, especially when reviewing contracts, summarizing specifications, checking bid documents, and identifying missing information. However, because the company performs specialized concrete work, AI tools may not always be accurate enough for final estimating decisions.

The second company is Estudios y Consultorías S.A.S., a financial consulting company that helps businesses understand costs, assets, liabilities, and financial performance. Even though this company is not a contractor, financial analysis is still important for AEC-related decisions because companies need to understand project costs, investments, and business performance. AI could help in this type of work by organizing financial information, summarizing reports, identifying patterns, and preparing drafts. However, because financial data is sensitive, AI also creates concerns related to confidentiality, accuracy, data protection, and professional responsibility (European Central Bank, 2024).

The purpose of this report is to compare how AI can be used in these two different company settings. The goal is to identify the main opportunities, barriers, and risks of AI adoption. This report does not treat AI as something that replaces professionals; instead, it looks at AI as a tool that can help people work more efficiently when it is used carefully. In both construction estimating and financial consulting, AI can be valuable, but it still requires human review, training, trust, and a clear understanding of where it actually helps.

Literature Review

Generative artificial intelligence has changed the way many organizations think about productivity and information management. McKinsey & Company (2023) explains that generative AI has expanded quickly because it can support tasks such as writing, summarizing, analyzing information, and improving workflow efficiency. This is important for professional industries because many employees now have the access to AI tools that can help with daily tasks without requiring advanced technical knowledge. Instead of only being used for highly technical applications, AI is now being tested in areas such as communication, report, document review, and decision support.

In AEC-related work, information management is one of the areas where AI can provide value. Construction and engineering projects usually include contracts, drawings, specifications, addendums, schedules, reports, RFIs, and other documents that must be reviewed carefully. Abioye et al. (2021) found that AI has been applied in construction for activities such as risk management, activity monitoring, resource optimization, and project decision-making. These applications show that AI can support companies by helping process large amounts of project information and identifying issues that may affect cost, schedule or performance.

Generative AI is also becoming more relevant in construction because many project tasks involve text-based information. Alwashah et al. (2025) discuss how generative AI can be applied in construction use cases and future workflows. For example, generative AI can help summarize project documents, support communication, organize technical information, and assist with first drafts of reports. This is useful in AEC because many project decisions depend on understanding written requirements and communicating them clearly. However, the literature also suggests that these tools should be used carefully because AI-generated summaries may not always capture the full context of a project.

A major topic in the literature is that AI adoption depends on more than just access to technology. Regona et al. (2022) found that construction companies face adoption challenges related to cost, training, data quality, trust, and organizational readiness. These challenges are important because companies may see the value of AI but still struggle to implement it correctly. For specialized contractors, this issue can even be stronger because general AI tools may not understand field conditions, specific scopes, or the exact details of each particular project. Because of this, AI adoption in AEC requires both technology and experienced human review.

The literature on financial AI adoption also shows similar concerns. The European Central Bank (2024) explains that AI can support financial work by improving efficiency and helping organizations analyze information, but it can also create risks if not managed properly. Financial work depends on sensitive data, accurate numbers, and professional responsibility. Because of this, AI tools must be used with controls that protect confidential information and reduce the risk of incorrect outputs.

The U.S. Department of Treasury (2024) also discusses the risk associated with AI in financial services, including data privacy, model risk, bias, and dependence on third party providers. These issues are important for consulting companies because consultants often work with private client information. If employees enter sensitive data into AI tools without clear rules, the company could create confidentiality or data protection problems. This shows that AI adoption is not only a productivity issue, but also a governance issue.

The Financial Stability Board (2024) adds that AI use in financial services create concerns related to operational dependency, cyber risk, data gaps, and model governance. These concerns show that companies need policies and review procedures before relying heavily on AI tools. Even when AI produces useful summaries or analysis, professionals still need to verify the information and understand the limits of the tool.

Overall, the literature shows that AI can improve efficiency and support decision-making, but successful adoption depends on a variety of factors. This connects directly to the companies studied in this report because both construction estimating and financial consulting involve sensitive information and decisions that require accuracy. AI can be valuable, but the literature supports the idea that it should be used as a support tool rather than a replacement for professional judgment.

Methodology

This project used a qualitative research approach based on two professional interviews. The purpose of this method was to collect practical insight into how AI is being used or considered in real company settings connected to AEC-related work.

Two professionals were selected for the interviews: Juan Romero, Lead Estimator at Gosalia Concrete Constructors, Inc., and Diego Rapalino, Financial Consultant at Estudios y Consultorías S.A.S. These participants were selected because their roles provided different perspectives on AI adoption: one from a construction estimating standpoint and the other from a financial consulting perspective.

Both interviews were conducted by phone. Before the interviews, a list of structured questions was prepared and used with both participants. The questions focused on AI tools, possible uses of AI, benefits, barriers, risks, and future expectations. Using the same general questions for both interviews helped keep the responses organized and made it easier to compare the findings.

After the interviews, the responses were reviewed and grouped into common themes. The main themes included AI as a support tool, the need for human review, concerns about accuracy, data privacy, training, and the difficulty of finding AI tools that fit specific company needs. These themes were then used to compare both companies and connect the interview findings with the academic literature.

Findings and Thematic Discussion

The interview responses showed that both companies see AI as a useful support tool, but not as a replacement for professional judgment. In both interviews, AI was discussed as something that can help organize large amounts of information, summarize documents, identify patterns, and support decision-making. However, both professionals also emphasized that AI-generated information still needs to be reviewed by a person with experience in the specific type of work being performed.

A comparison of the interview responses is shown in Table 1.

Table 1

Comparison of Interview Findings

Company	Role Interviewed	AI Tools or Uses Mentioned	Main Benefits	Main Barriers or Risks
Gosalia Concrete Constructors, Inc	Lead Estimator	ChatGPT and similar tools for contract review, specification summaries, bid document review, and identifying missing or conflicting information	Saves time reviewing long documents, support estimating, helps organize risks before bidding	Accuracy, trust, human review, project-specific condition, lack of AI tools made for specialized concrete work
Estudios y Consultorias S.A.S.	Financial Analyst	ChatGPT, Microsoft Copilot, Gemini, and AI features in Excel for financial summaries, report drafting, data review and identifying patterns	Saves time, improves organization, helps explain financial information, supports report preparation	Confidentiality, data protection, accuracy, professional responsibility, need to verify financial information

- ***AI as a Support Tool***

One of the strongest topics from both interviews was that AI is useful as a support tool. At Gosalia Concrete Constructors, Inc., AI was seen as helpful for reviewing contracts, summarizing specifications, checking bid documents, etc. These tasks can take a lot of time during the estimating process, especially when projects include long scopes of work, detailed specifications, addendums, or special requirements.

At Estudios y Consultorías S.A.S., AI was also described as useful for support tasks, but in a different way. In financial consulting, AI can help organize financial information, summarize reports, identify patterns, and prepare drafts. This shows that AI can help different types of companies, but its value depends on the type of work being performed. In both interviews, AI was not presented as the final decision-maker, but as a support tool to make work more efficient.

- ***Importance of Human Review***

Another major theme that both of the professionals mentioned was the need for human review. In the construction estimating interview, the concern was that AI tools may not fully understand the details of specialized concrete work. A tool may be able to summarize a contract of specifications, but it may not correctly understand project-specific conditions, production issues, traffic restrictions, access problems, or bid risks. Because of this, the estimator still needs to review the information before making any final decisions.

In the financial consulting interview, human review was also important because financial information must be accurate and confidential. AI may help organize data, prepare a summary, or even help with some mathematical information, but financial professionals still have to verify the numbers, conclusions, and recommendations. This finding connects with the idea that AI can improve productivity, but it still needs professional judgment to avoid mistakes.

- ***Accuracy and Trust***

Accuracy and trust were also common concerns in both interviews. For Gosalia Concrete Constructors, Inc., accuracy is important because estimating mistakes can affect the bid price, project costs, and company risk. If AI misses a requirement or misunderstands a specification, the company could make a bad decision during the bidding process. This is one reason why AI tools are useful, but they can't be trusted blindly.

For Estudios y Consultorías, S.A.S., accuracy is also a major concern because financial reports and cost studies affect business decisions. If AI produces incorrect summaries or wrong interpretations of key information, the final recommendations could be totally unreliable. Both interviews showed that trust in AI depends on how well the tool performs and how carefully the results are checked.

- ***Difficulty Finding Industry-Specific AI Tools***

A specific finding from the Gosalia interview was that the company has not found an AI tool that works perfectly for its specialized concrete work. Many tools are general or focused on broader construction management, but they may not be accurate enough for roadway barrier walls, concrete pavement, slab replacement, and similar scopes. This creates a barrier because the company may see value in AI but still struggle to find a tool that fits exactly their needs.

This issue was less specific in the financial consulting interview, where general tools like ChatGPT, Copilot, Gemini, and Excel AI features were seen as more directly useful for writing, organizing, and summarizing information. This difference shows that AI adoption may be easier in tasks that involve text, reports, and data organization rather than highly specialized estimating tasks that depend on field conditions and technical judgement.

- ***Training and Future Adoption***

Both interviews also showed that training is important for future AI adoption. AI tools can be helpful, but employees need to understand how to use them correctly, what information should not be entered, and how to check the results. Without training, there is a higher risk of mistakes, misuse, or overreliance on AI-generated information.

Overall, both professionals suggested that AI adoption will likely continue to grow. In construction estimating, AI may become more common for bid support, scheduling and project management. In financial consulting, AI may become more common for reports, data analysis, financial summaries, and client presentations. However, in both settings, AI adoption will depend on trust, training, data protection, and the ability to use the tool with responsibility.

Discussion, Conclusion, and Recommendations

- ***Discussion***

The findings from this project show that AI can create value in both construction estimating and financial consulting, but only when it is used carefully. In both interviews, AI was viewed as a tool that can support professionals by saving time, organizing information, and helping with document review or data analysis. However, the responses also showed that AI should not be used as a replacement for professional experience. The most important point from both companies is that AI can help with the first steps of a task, but the final review and final decision still need to come from a trained professional.

For Juan Romero, from Gosalia Concrete Constructors, Inc., AI adoption is useful but limited by the specialized nature of the company’s work. General AI tools can help review some contracts, summarize specifications, check bid packages, and find missing or wrong information. However, estimating concrete scopes such as pavements, barrier walls, and slab replacements require project specific knowledge. Field conditions, production rates, traffic restrictions, and access issues are not always easy for AI to understand. Because of this, AI can support the estimating team, but it can’t replace the estimator’s judgment.

For Diego Rapalino, from Estudios y Consultorias S.A.S., AI appears to be easier to apply because much of the work involves reports, financial information, summaries, and data organization. Tools such as ChatGPT, Copilot, Gemini, and AI features in Excel can help prepare drafts and improving the report. However, the main concern in this environment is confidentiality and accuracy. Since financial consulting deals with sensitive information, AI must be used carefully to protect data and avoid incorrect conclusions.

Overall, the comparison shows that AI adoption depends heavily on the type of work being performed. AI may be easier to use in tasks involving text, reports, and summaries, and it may be more difficult to use in specialized technical work where accuracy, field experience, and project specific conditions are very important. This connects with this literature because AI adoption is not only about having access to the technology, it also depends on trust, training, data quality, and proper human review.

- ***Recommendations***

Based on the findings, one recommendation is that companies should begin using AI in low-risk tasks before applying it to critical decisions. For example, AI can be used to summarize long documents, draft internal notes, or prepare first drafts of reports. These tasks can save time without giving AI full control over final decisions.

A second recommendation is that companies should create clear rules for AI use. Employees should know what type of information can and can’t be entered into AI tools. This is especially important when dealing with bid pricing, internal production rates, contracts, financial statements, client information, and other confidential data.

A third recommendation is to provide basic AI training for employees. Training should focus on how to write effective prompts, how to check AI-generated information, and how to identify mistakes or incomplete answers from AI. Employees should understand that AI can be helpful, but it still needs to be reviewed carefully.

- ***Conclusions***

In conclusion, AI has strong potential to support companies connected to the AEC industry, but its value depends on how it is used. In construction estimating, AI can help with bid support and document review, but it still needs human judgment because every project has different type of conditions. On the other hand, in financial consulting, AI can help organize data, summarize reports, and support analysis, but confidentially and accuracy must be carefully protected.

The main takeaway from this project is that AI should be treated as a support tool, not as a replacement for professionals. When used responsibly, AI can help companies work more efficiently and faster. However, successful AI adoption requires training, trust, data protection, and last but not least, human review. For both companies studied, the best approach is to use AI carefully in tasks where it adds value while keeping final decisions in the hands of experienced professionals.

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Appendix A

Interview Questions

Can you briefly explain your role in the company?

How is AI currently being used, or how could it be used, in your type of work?

What type of AI tools or AI-supported software do you normally use or see being useful?

In what tasks do you think AI could save time or improve productivity?

What are the main benefits of using AI in your company or industry?

What are the main barriers or concerns related to AI adoption?

How important is human review when using AI-generated information?

Do you think AI could replace professionals in your field, or is it more of a support tool?

What risks should companies consider before using AI tools?

How do you think AI adoption will change in the next few years?

Appendix B

Interview Notes

Interview 1: Juan Romero, Lead Estimator from Gosalia Concrete Constructors, Inc.

<u>Question 1: Can you briefly explain your role in the company?</u>

Juan Romero explained that his role as Lead Estimator involves reviewing plans, specifications, quantities, and project requirements to prepare bids.

<u>Question 2: How is AI currently being used, or how could it be used, in your type of work?</u>

AI could be useful for reviewing bid documents, summarizing specifications, identifying important requirements, and comparing information between drawings and proposal documents.

<u>Question 3: What type of AI tools or AI-supported software do you normally use or see being useful?</u>

Juan explained that the company has not found an AI tool that works perfectly for its specific concrete scopes. However, tools like ChatGPT can help review and summarize contracts, specifications, and bid documents.

<u>Question 4: In what tasks could AI save time or improve productivity?</u>

AI could save time by summarizing long specifications, reviewing addendums, drafting scope clarifications, organizing bid questions, preparing emails, and summarizing project risks.

<u>Question 5: What are the main benefits of using AI in your company or industry?</u>

The main benefits are saving time, organizing information faster, and helping estimators identify important project requirements before submitting a bid.

<u>Question 6: What are the main barriers or concerns related to AI adoption?</u>

The biggest concerns are accuracy, trust, and the lack of AI tools designed for specialized concrete estimating.

Question 7: How important is human review when using AI-generated information?

Human review is very important because estimating depends on project-specific details, field conditions, and accurate information.

Question 8: Do you think AI could replace professionals in your field, or is it more of a support tool?

AI is more of a support tool. It can help with repetitive tasks, but estimating still requires experience, judgment, and knowledge of field conditions.

Question 9: What risks should companies consider before using AI tools?

Companies should consider risks related to data privacy, bid pricing, internal production rates, contracts, and other sensitive project information.

Question 10: How do you think AI adoption will change in the next few years?

AI will likely become more common for document review, bid support, scheduling, and project management, but it will still require careful review.

Interview 2: Diego Rapalino, Financial Analyst for Estudios y Consultorias S.A.S.

Question 1: Can you briefly explain your role in the company?

Diego Rapalino explained that his work focuses on financial consulting, cost analysis, assets, liabilities, and helping companies understand their financial position.

Question 2: How is AI currently being used, or how could it be used, in your type of work?

AI could be useful for organizing financial information, summarizing reports, identifying trends, and preparing first drafts of analysis.

Question 3: What type of AI tools or AI-supported software do you normally use or see being useful?

Tools like ChatGPT, Microsoft Copilot, Gemini, and AI features in Excel could be useful for summarizing financial documents, drafting reports, creating tables, and identifying patterns.

Question 4: In what tasks could AI save time or improve productivity?

AI could save time in report writing, document summaries, financial data organization, first drafts of analysis, and presentation preparation.

Question 5: What are the main benefits of using AI in your company or industry?

The main benefits are saving time, improving organization, supporting report preparation, and helping consultants focus more on interpreting information.

Question 6: What are the main barriers or concerns related to AI adoption?

The main concerns are confidentiality, accuracy, data protection, and depending too much on AI-generated information.

Question 7: How important is human review when using AI-generated information?

Human review is very important because financial work requires accurate numbers, correct conclusions, and professional responsibility.

Question 8: Do you think AI could replace professionals in your field, or is it more of a support tool?

AI is more of a support tool. It can help with repetitive tasks and summaries, but consultants still need professional judgment and experience.

Question 9: What risks should companies consider before using AI tools?

Companies should consider confidentiality, incorrect outputs, data privacy, and the use of outside AI platforms with sensitive client information.

Question 10: How do you think AI adoption will change in the next few years?

AI will likely become more common in financial consulting for document review, report writing, data analysis, financial summaries, and client presentations.